

SMART REAL-TIME AGE, GENDER, EMOTION, AND BEHAVIOR-BASED ADVERTISING SYSTEM FOR SRI LANKAN MALLS

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Final Report

BSc (Hons) degree in Information Technology

Specializing in Data Science

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Sri Lanka Institute of Information Technology

Sri Lanka

April 2026

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Dissertation was submitted in partial fulfilment of the requirements for the B.Sc.
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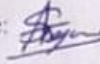
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DECLARATION

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ABSTRACT

The increasing adoption of digital signage and intelligent retail technologies has created a growing demand for advertising systems capable of understanding audience behavior in real time. Conventional advertising displays in shopping malls typically present static or manually scheduled content without incorporating context-sensitive audience analytics. This limits their ability to adapt advertisements according to viewer characteristics, engagement levels, and behavioral patterns observed in the environment. In response to this limitation, this project presents a real-time context-aware behavior-driven engagement analytics framework for advertising systems in Sri Lankan malls.

The proposed system integrates computer vision, lightweight deep learning, and behavioral intelligence to monitor human interaction with a public display and support adaptive advertisement delivery. The solution consists of a dual-output architecture that combines perception-level inference with higher-level engagement analytics. At the perception level, the system employs person detection, multi-object tracking, head orientation estimation, and motion analysis to extract low-level behavioral signals from live video streams. At the intelligence level, these signals are transformed into session-based engagement indicators using contextual reasoning, spatial zone awareness, temporal adaptation, and historical behavioral statistics. The system further supports long-term analytics through data logging, clustering of audience behavior, and time-series forecasting of engagement trends.

A custom lightweight convolutional neural network is used as part of the real-time perception pipeline to support efficient inference in resource-constrained deployment conditions. The overall framework is designed with practical deployment feasibility in mind, considering latency, hardware limitations, environmental variability, and scalability for real-world mall environments. Experimental evaluation demonstrates that the proposed approach improves the stability of engagement estimation, reduces false alarms compared to basic frame-level methods, and enables meaningful behavioral segmentation and future engagement forecasting.

The findings indicate that context-aware engagement analytics can significantly enhance the value of intelligent advertising systems in mall environments by supporting more relevant content delivery, better audience understanding, and stronger commercialization potential. The project contributes to the intersection of data science, artificial intelligence, and real-time system design by demonstrating how behavioral analytics can be operationalized in a practical retail setting.

Keywords: engagement analytics, intelligent advertising, digital signage, real-time computer vision, lightweight CNN, behavioral clustering, smart retail, Sri Lankan malls

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to our supervisor, Dr. Nathalie Silva, for her continuous guidance, invaluable feedback, and encouragement throughout the duration of this research project. Her support has been instrumental in shaping this study from its inception to completion. I also extend my sincere appreciation to my co-supervisor, Mrs. Shashini Kumarasinghe, for her consistent support and insightful input throughout the project. Lastly, I am deeply grateful to my teammates, friends, and family for their support, encouragement, and understanding throughout this journey.

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INTRODUCTION

The retail industry is undergoing rapid digital transformation as businesses increasingly seek data-driven methods to improve customer experience, operational efficiency, and revenue generation [1], [2]. Among the technologies shaping this transition, digital advertising displays and intelligent signage systems have emerged as important tools for customer engagement in commercial spaces. In large shopping malls, such displays are commonly used to promote products, communicate brand messages, and influence customer decisions at points of interaction. However, despite their visual prominence, many existing display systems remain passive, static, and disconnected from the behavior of the audience observing them [3].

Traditional advertising screens operate using predefined playlists, fixed schedules, or manual content rotation strategies. Although such approaches are simple to implement, they lack responsiveness to the real-time context of the viewing environment [4]. In high-traffic mall settings, customer interactions with displays are dynamic and highly variable. People differ in age group, gender, emotional state, movement behavior, and the extent to which they visually attend to digital content [5]. As a result, a one-size-fits-all content strategy is often inefficient, leading to low engagement and missed personalization opportunities.

With the advancement of artificial intelligence, computer vision, and edge-based deep learning, it has become increasingly feasible to build systems capable of understanding human behavior directly from video streams [6], [7]. Vision-based audience analytics can now support applications such as people counting, demographic estimation, facial expression recognition, gaze approximation, motion tracking, and dwell analysis. When combined in an intelligent framework, these capabilities can transform passive advertising displays into adaptive systems that respond to the surrounding audience.

In the Sri Lankan context, shopping malls represent a particularly relevant environment for such systems. Urban retail spaces in Sri Lanka are becoming more technologically sophisticated, yet intelligent advertising remains relatively underexplored compared to international smart retail ecosystems. Mall operators and advertisers continue to depend largely on conventional promotional strategies without deep integration of behavioral analytics. This creates an opportunity for locally relevant research that bridges real-time artificial intelligence and commercial advertising use cases within Sri Lankan public retail environments [8].

The present project addresses this need by proposing a real-time context-aware behavior-driven engagement analytics system for advertising in Sri Lankan malls. The system is designed to observe audience behavior through a live camera feed, extract behavioral signals, estimate engagement levels, and support adaptive advertisement delivery. Rather than relying solely on instantaneous frame-level observations, the framework introduces session-based behavioral

reasoning, contextual adjustment, and long-term analytics. This allows the system to move beyond simple detection and toward intelligent behavioral interpretation.

The project is situated at the intersection of data science, artificial intelligence, computer vision, and real-time systems engineering. From a data science perspective, the system depends on continuous data acquisition, preprocessing, behavioral feature construction, clustering, forecasting, and metric-based evaluation [9]. From an AI perspective, it leverages lightweight deep learning models for real-time perception. From a systems perspective, it emphasizes low-latency operation, modular architecture, and deployment feasibility. This multidisciplinary combination makes the project academically relevant while also supporting practical commercialization pathways in retail technology.

1.1 Background & Literature Review

Audience measurement and engagement analysis have been longstanding topics in the fields of human-computer interaction, digital signage, and smart retail [10] , [11]. Early studies in public display systems primarily investigated whether users noticed or interacted with a display at all. These approaches frequently depended on direct observation, manual annotation, or simple proximity-based sensing. While useful as proof of concept, they lacked scalability and could not provide nuanced insight into the behavioral patterns of different viewers.

Visual Attention Estimation Using Computer Vision

As computer vision matured, public display research began incorporating automated visual attention estimation techniques [12]. Gaze detection, head pose analysis, and dwell-time heuristics became common mechanisms for approximating whether a viewer was engaged with a display. Gaze-based methods, in theory, provide a strong signal of visual attention. However, accurate gaze estimation in public environments remains technically challenging because of camera placement constraints, variable lighting, partial occlusions, and the fact that viewers do not typically stand in ideal frontal positions [13]. For this reason, many practical systems use head orientation as a lower-cost proxy for visual attention.

Limitations of Dwell-Time-Based Engagement Metrics

Dwell-time analysis has also been widely used in retail analytics, museum installations, and surveillance-inspired behavioral studies. The underlying assumption is that individuals who remain near a display for longer periods are more likely to be interested in it.

Although dwell time is an intuitively meaningful feature, it is not sufficient on its own. In crowded mall environments, a person may dwell near a display due to congestion, waiting, or unrelated social behavior rather than actual engagement with the displayed content [14]. This introduces false positive cases that weaken the reliability of naive dwell-based systems.

Multi-Cue Engagement Modeling in Smart Retail

Recent work in smart retail has expanded beyond single-feature engagement models to include multi-cue approaches [15]. These systems often combine people's detection, tracking, demographic inference, motion analysis, and attention cues to produce richer representations of audience behavior.

Convolutional neural networks have been widely adopted for perception tasks due to their strong performance in image classification and object understanding [6]. Similarly, modern one-stage detectors such as YOLO have become attractive for real-time deployment because they provide a balance between accuracy and computational efficiency [16]. Multi-object tracking methods such as SORT further extend these capabilities by enabling persistent identity assignment across frames, allowing temporal analysis of behavior [17].

Spatial and Behavioral Analytics in Retail Environments

In parallel, the broader literature on intelligent retail and surveillance analytics has demonstrated the value of trajectory analysis, heatmaps, zone analytics, customer flow analysis, and behavioral segmentation [18]. Retail environments have motivated strong interest in understanding how people move through space, how long they remain in certain areas, and how this relates to marketing effectiveness.

However, many existing systems remain focused on aggregate counting or spatial occupancy statistics rather than individualized engagement interpretation. As a result, they provide descriptive analytics but not necessarily context-aware behavioral intelligence.

Another important line of work relates to unsupervised learning and time-series analytics. Clustering methods such as k-means have been used to discover recurring behavioral styles in mobility data, surveillance logs, and customer journey analysis [9]. Likewise, forecasting methods such as ARIMA and recurrent neural models have been used to predict demand, traffic, or human activity patterns over time [19].

Yet these analytical approaches are often disconnected from real-time perception systems. They are typically applied offline after data collection rather than being integrated into a unified intelligent advertising framework.

Privacy and Ethical Considerations

The use of biometric and behavioral data in public advertising introduces privacy challenges, especially in jurisdictions with strict data protection laws such as GDPR [20]. Public backlash against facial recognition in advertising campaigns has further emphasized the need for privacy-by-design solutions. Techniques such as anonymized skeletal tracking, on-device processing, and immediate feature extraction without storage have been recognized as best practices for ensuring compliance and maintaining consumer trust.

This system will allow Sri Lankan shopping malls to transition from static advertising to responsive, context-aware marketing, improving customer experience and increasing the return on investment for advertisers.

Deployment Challenges in Real-Time Systems

From a system design perspective, many prior studies face a trade-off between model complexity and deployment practicality. High-capacity deep learning models may achieve strong predictive performance in laboratory datasets but are difficult to deploy in resource-constrained real-time environments without GPUs or specialized hardware [21].

In commercial mall deployments, low-latency inference, robustness to environmental noise, and maintainability are often more important than maximizing benchmark accuracy under ideal conditions. This highlights the need for efficient, modular, and deployment-oriented designs.

The literature therefore suggests that, while major progress has been made in person detection, attention estimation, retail analytics, and predictive modeling, a gap remains in combining these elements into a context-aware real-time advertising intelligence system suited to practical deployment. The present study contributes to this need by integrating lightweight perception, session-based engagement modeling, contextual adaptation, and long-term behavioral analytics within a single architecture tailored to mall advertising environments.

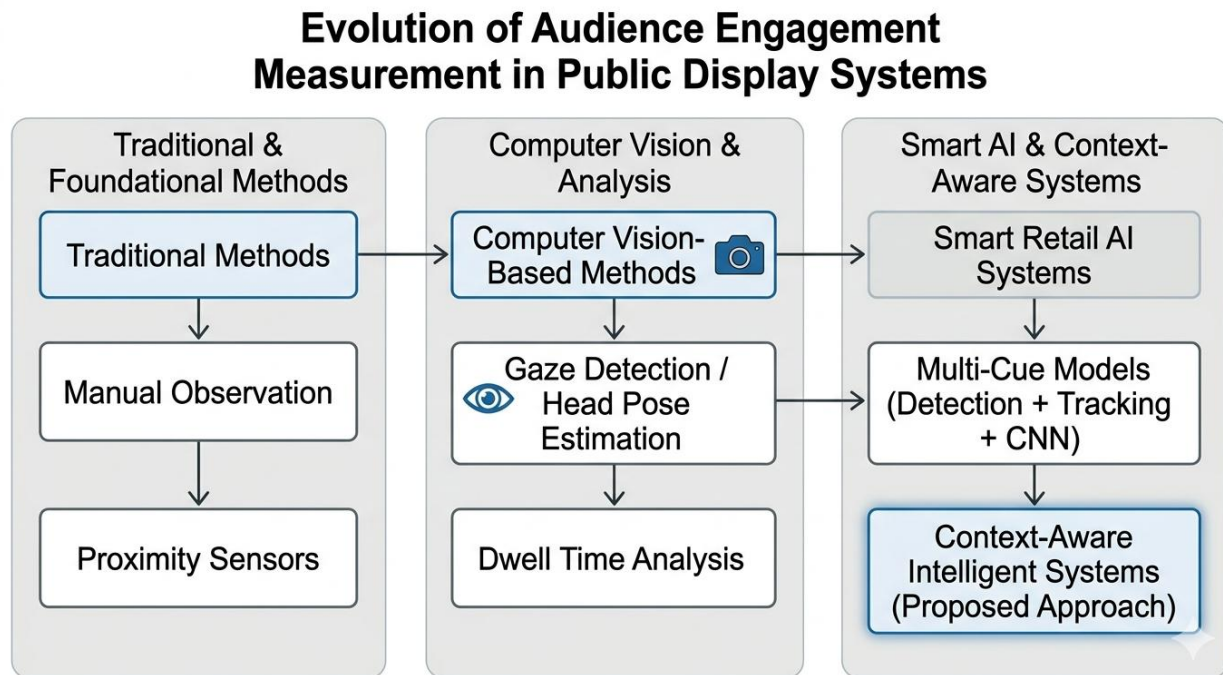


Figure 1: Evolution of Audience Engagement Measurement Techniques

1.2 Research Gap

Table 1: Comparison between existing systems

Feature	Existing Systems (Research)	Proposed Application Component
Engagement Modeling Approach	Engagement is typically estimated at the frame level using instantaneous cues such as gaze or head pose. These approaches are highly sensitive to noise, transient gaze shifts, and short-term occlusions [12].	Engagement is modeled at the session level, where behavioral signals are aggregated over time. This reduces instability and provides a more reliable estimate of sustained user interaction.
Context Awareness	Most systems lack contextual interpretation. Identical behaviors are treated uniformly regardless of location, time, or environmental conditions [18].	Introduces context-aware modeling, including spatial zone reasoning, time-of-day sensitivity, and crowd-density-aware adjustments for accurate interpretation.
Spatial Intelligence (Zone Awareness)	Minimal or no use of spatial segmentation. Engagement decisions are not influenced by the physical layout of the environment [18].	Utilizes semantic zone mapping (interaction zones, walkways, entry areas) with zone-based weighting to refine engagement estimation.
Use of Detection & Tracking Models	Detection and tracking are primarily used for counting, tracking, or visualization, with limited transformation into higher-level behavioral intelligence [17].	Integrates YOLO detection and SORT tracking outputs into behavior-driven analytics, enabling extraction of dwell time, motion patterns, and engagement signals.
Behavioral Interpretation	Focuses on descriptive analytics such as people counting, heatmaps, or trajectory visualization rather than engagement understanding [18].	Converts low-level features into behavioral intelligence, including engagement scoring, anomaly detection, and user interaction classification.
Data Utilization Strategy	Real-time data is often used only for immediate inference and then	Implements MongoDB-based session logging, enabling structured storage of behavioral

	discarded. Limited use of historical behavioral data [9].	data for long-term analytics and system improvement.
Longitudinal Learning Capability	Lack of long-term learning mechanisms. Systems do not adapt based on historical patterns or deployment conditions [19].	Supports deployment-adaptive learning, where thresholds and behavioral interpretations evolve using historical data.
Anomaly Detection	Basic anomaly detection, often based on simple thresholding without context or persistence logic.	Uses context-aware anomaly detection, combining dwell, motion, zone constraints, and temporal persistence to reduce false alerts.
Robustness to Real-World Conditions	Performance degrades under crowd congestion, occlusion, and environmental variability due to static thresholds [21].	Uses crowd-density-aware adaptive thresholds, improving robustness in dynamic mall environments.
System Architecture	Often single-layer systems focusing only on perception or analytics independently.	Implements a dual-output architecture combining real-time inference with offline behavioral intelligence.
Computational Efficiency	Many systems rely on heavy models requiring high-end hardware, limiting practical deployment [6].	Utilizes a lightweight CNN and optimized pipeline, ensuring real-time performance on moderate hardware.

The comparison presented in the table highlights a clear distinction between conventional research systems and the proposed application framework. Existing approaches largely emphasize low-level perception and short-term inference, often relying on frame-based engagement estimation and context-independent analysis [12]. While these methods are effective for basic detection and visualization tasks, they fall short in capturing the complexity of real-world human behavior in dynamic retail environments. In contrast, the proposed system adopts a more holistic and application-oriented perspective by integrating session-level engagement modeling, contextual awareness, and spatial intelligence into a unified framework. This enables more stable, interpretable, and meaningful representations of audience behavior.

Furthermore, the proposed architecture bridges the gap between real-time analytics and long-term learning by incorporating structured data storage and historical analysis capabilities. Unlike traditional systems that discard behavioral data after immediate processing, this approach

facilitates the accumulation of knowledge over time, allowing the system to adapt to environment-specific patterns and improve its decision-making processes [9]. The inclusion of clustering, anomaly detection, and predictive modeling further extends the system beyond descriptive analytics toward intelligent, behavior-driven insights [19].

Importantly, the design also considers practical deployment constraints, particularly within emerging retail contexts such as Sri Lanka. By prioritizing computational efficiency, modularity, and robustness to environmental variability, the system ensures feasibility in real-world mall settings without requiring specialized infrastructure [21]. Overall, the proposed solution addresses critical limitations identified in literature and advances the state of the art by transforming vision-based perception into a context-aware, adaptive, and commercially viable advertising intelligence system.

1.3 Problem Statement

Conventional digital advertising systems in shopping malls are largely static and non-responsive to the actual behavior of nearby audiences. Even when basic analytics are incorporated, these systems typically rely on simplistic engagement measures such as proximity detection or instantaneous gaze estimation [12]. While these signals provide a rough indication of attention, they are insufficient for reliable behavioral understanding in complex, real-world environments. Such approaches are highly sensitive to noise, temporary occlusions, lighting variations, and abrupt movements, leading to false detections and inconsistent engagement assessment [13]. As a result, the insights generated are often shallow and lack practical interpretability for decision-making.

In a mall setting, customer behavior is inherently dynamic and influenced by multiple contextual factors, including movement flow, spatial positioning within the environment, crowd density, and temporal variations such as peak and off-peak hours [18]. A system that fails to incorporate these contextual dimensions cannot accurately differentiate between genuine engagement and incidental presence. For example, a person standing near a display may be waiting, passing through congestion, or interacting with others rather than engaging with the advertisement. The inability to distinguish such scenarios significantly reduces the effectiveness of advertising strategies and limits the system's ability to deliver meaningful audience insights.

Moreover, existing systems predominantly operate on short-term inference without leveraging historical behavioral data. The absence of structured session logging and longitudinal analysis prevents the identification of recurring audience patterns, behavioral segmentation, and evolving engagement trends over time [9]. Without this capability, systems remain reactive—responding only to immediate inputs—rather than proactive or adaptive. This limitation restricts opportunities for optimizing advertisement delivery, personalizing content, and supporting data-driven commercial strategies.

Another critical challenge lies in the gap between low-level perception and high-level behavioral intelligence. Although modern computer vision techniques enable accurate detection and tracking of individuals, these outputs are often underutilized, serving primarily for counting or basic trajectory visualization [17]. The transformation of these raw signals into meaningful engagement metrics, anomaly detection, and predictive insights remains insufficiently explored, particularly in the context of real-time advertising systems.

Additionally, many existing solutions are developed in controlled or resource-rich environments and assume the availability of high-end computational infrastructure, extensive labeled datasets, and stable operating conditions [21]. Such assumptions are not always feasible in emerging retail contexts like Sri Lanka, where cost constraints, environmental variability, and deployment scalability must be carefully considered. This creates a significant gap between research prototypes and practical, real-world implementations.

Therefore, the core problem addressed in this research is the absence of a scalable, real-time, and context-aware engagement analytics framework that can operate effectively in dynamic mall environments. The system must go beyond frame-level analysis to incorporate session-based behavioral modeling, integrate contextual awareness, support anomaly-aware reasoning, and enable long-term data-driven intelligence. Addressing this problem is essential for developing adaptive advertising systems that not only observe audience behavior but also learn from it, evolve over time, and provide actionable insights tailored to the Sri Lankan retail landscape.

1.4 Objectives

1.4.1 Main Objective

The primary aim of this research is to design, develop, and evaluate a real-time, context-aware, behavior-driven engagement analytics system tailored for intelligent advertising in Sri Lankan mall environments. The system leverages advances in artificial intelligence, computer vision, and data-driven behavioral intelligence to move beyond traditional static advertising toward adaptive, audience-aware content delivery.

This research seeks to bridge the gap between low-level visual perception and high-level behavioral understanding by integrating real-time analytics with long-term learning capabilities. In doing so, it aims to provide a scalable and deployment-ready solution that not only detects and tracks individuals but also interprets their behavior within contextual constraints, enabling more accurate engagement estimation and actionable commercial insights.

1.4.2 Specific Objectives

- To design and implement a real-time perception pipeline capable of accurately detecting, tracking, and behaviorally analyzing individuals interacting with an advertising display.

This includes integrating object detection and multi-object tracking techniques to ensure continuous identity preservation across frames.

- To develop a lightweight and computationally efficient model architecture that balances accuracy and performance, ensuring low-latency inference and feasibility for deployment on moderate hardware commonly available in mall environments.
- To construct a robust engagement estimation mechanism by combining multiple behavioral cues such as visual attention (e.g., head orientation), dwell time, and motion dynamics, enabling a more comprehensive and reliable representation of user interaction.
- To enhance engagement reliability through contextual intelligence, incorporating spatial zone awareness, crowd-density-sensitive thresholding, and session-level behavioral aggregation to reduce noise and false positives in real-world scenarios.
- To design and implement a session-based engagement modeling framework, where user interactions are analyzed over time rather than at isolated frames, improving stability and interpretability of engagement predictions.
- To implement a dual-output system architecture, enabling both real-time advertising response (for immediate content adaptation) and persistent behavioral logging (for offline analytics and system improvement)
- To develop a structured data storage mechanism using a database system (e.g., MongoDB) to record interaction sessions, supporting efficient retrieval, analysis, and long-term knowledge accumulation.
- To apply unsupervised learning techniques such as clustering to identify recurring audience behavior patterns, enabling segmentation of users based on interaction styles and engagement characteristics.
- To evaluate the overall system performance using quantitative metrics such as classification accuracy, precision, recall, and latency, as well as qualitative analysis of behavioral insights and system outputs.
- To assess the practical applicability and commercialization potential of the proposed system within the Sri Lankan retail context, considering factors such as cost-efficiency, scalability, robustness, and ease of deployment.

METHODOLOGY

This chapter presents the methodological framework adopted for the design, development, and evaluation of the proposed real-time context-aware engagement analytics system for advertising in Sri Lankan mall environments. The methodology integrates principles from artificial intelligence, data science, and real-time system engineering to transform raw visual data into meaningful behavioral insights [9].

The proposed system follows a structured approach that begins with requirement analysis and proceeds through system architecture design, data collection, preprocessing, model development, and implementation. A key aspect of the methodology is the separation of the system into real-time operational components and offline analytical components, allowing the framework to simultaneously support low-latency decision-making and long-term behavioral intelligence.

The methodology is also designed with practical deployment considerations in mind. Factors such as computational efficiency, scalability, hardware constraints, and adaptability to Sri Lankan retail environments have been carefully considered throughout the system design. A lightweight model architecture is employed to ensure real-time performance without requiring high-end computational resources [21].

Furthermore, a comprehensive testing and evaluation strategy is adopted to assess system performance across multiple dimensions, including classification accuracy, real-time responsiveness, robustness under varying crowd conditions, and the effectiveness of behavioral analytics components [22].

2.1 Requirement Gathering and Analysis

The development of the proposed real-time engagement analytics system began with a structured and systematic requirement gathering and analysis phase. This phase played a foundational role in shaping the overall system architecture by identifying key objectives, constraints, and performance expectations. It ensured that the proposed solution was not only technically sound but also aligned with real-world deployment conditions and commercial needs within Sri Lankan mall environments.

To achieve this, multiple sources of insight were considered. A comprehensive review of existing literature on intelligent advertising, computer vision-based engagement detection, and smart retail analytics was conducted to understand the current state of the art and its limitations. In parallel, existing commercial and research systems were analyzed to identify practical gaps in functionality, scalability, and contextual adaptability.

The requirements were derived based on three key perspectives to ensure a balanced and deployment-oriented system design:

- **Technical Requirements** focused on enabling robust real-time computer vision capabilities, including accurate person detection, multi-object tracking, and behavioral analysis under constraints such as limited computational resources, varying lighting conditions, and partial occlusions. The system must maintain low latency while ensuring acceptable levels of accuracy and stability.
- **Business Requirements** emphasized the need for actionable insights that support intelligent advertisement delivery. This includes reliable engagement estimation, audience behavior interpretation, and the ability to generate analytics that can inform marketing strategies, improve advertisement effectiveness, and support commercialization objectives.
- **Operational Requirements** addressed practical deployment considerations, including hardware affordability, ease of installation, scalability across multiple mall locations, and robustness to environmental variability such as crowd congestion, camera placement limitations, and dynamic indoor conditions.

Based on these requirements, the system was designed to operate continuously on live video streams, enabling real-time detection of human presence, tracking of individual movement, and analysis of behavioral patterns in proximity to advertising displays.

To effectively support both immediate responsiveness and long-term intelligence, dual-output architecture was adopted. This architecture allows the system to generate real-time engagement decisions for adaptive advertisement control while simultaneously storing structured behavioral session data for offline analytics, pattern discovery, and predictive modeling.

2.1.1 Functional Requirements

Functional requirements describe the specific operations and capabilities that the system must perform to achieve its intended purpose. The proposed system integrates computer vision, behavioral analytics, and data processing techniques to enable both real-time engagement detection and long-term behavioral intelligence. These requirements define how the system processes visual data, extracts meaningful insights, and supports adaptive advertising decisions.

Real-Time Human Detection

The system must be capable of detecting individuals in real time using a vision-based detection model. This functionality serves as the entry point of the perception pipeline and is fundamental to all subsequent processing stages.

The detection module must accurately identify all visible persons within each frame, even under challenging conditions such as varying illumination, background clutter, and partial occlusions. High detection accuracy is essential to ensure that no relevant subjects are missed, as errors at this stage propagate through the entire system.

Multi-Object Tracking

Following detection, the system must track multiple individuals across consecutive frames while maintaining persistent identities. This is achieved through a multi-object tracking mechanism that associates detections over time.

Continuous tracking is crucial for constructing temporal trajectories, which enable the computation of behavioral features such as dwell time and motion patterns. The tracking module must handle identity preservation even in crowded scenes where occlusions and overlapping individuals are common.

Attention Estimation via Head Orientation

The system must estimate head orientation for each detected individual to determine whether the person is visually attending the display. This approach serves as a practical proxy for gaze estimation in real-world environments where precise eye-tracking is not feasible.

By analyzing head pose, the system can infer attention direction with reasonable accuracy while maintaining computational efficiency. This component plays a critical role in distinguishing between passive presence and active engagement.

Behavioral Feature Extraction

A key functional requirement is the continuous computation of behavioral features for each tracked individual. These features include dwell time, motion magnitude, and looking ratio.

- Dwell time captures how long an individual remains within a specific area.
- Motion magnitude reflects movement intensity and behavioral dynamics.
- Looking ratio estimates the proportion of time an individual is visually oriented toward the display.

These features collectively form the foundation for engagement analysis and must be computed reliably over time.

Engagement Score Computation

Based on the extracted behavioral features, the system must calculate an engagement score using a predefined model. This score represents the likelihood that an individual is actively interacting with the display.

The scoring mechanism must integrate multiple behavioral signals in a balanced manner, reducing reliance on any single feature and improving robustness against noise and environmental variability.

Engagement Classification

The system must classify individuals as engaged or not engaged based on the computed engagement score and predefined thresholds.

This classification represents the primary output of the real-time decision layer. The classification results directly influence advertisement triggering logic and must therefore be both accurate and stable over time.

Anomaly Detection

In addition to engagement detection, the system must support anomaly detection. It should identify irregular or unexpected behaviors such as excessive dwell time, unusual motion patterns, or presence in restricted zones. This functionality enhances system reliability, supports security-related observations, and helps filter out misleading engagement signals.

Spatial Zone Assignment (Context Awareness)

The system must assign spatial zones to individuals based on their location within the scene. These zones may include interaction zones, walkways, and entry/exit areas. This significantly improves the accuracy of engagement estimation in complex environments.

Session-Based Data Logging

The system must log session-level behavioral data into a structured database. Each interaction session should include features such as dwell time, motion statistics, looking ratio, engagement score, and contextual attributes. This persistent storage enables long-term analysis, system evaluation, and continuous improvement of engagement models.

Behavioral Pattern Analysis (Clustering)

The system must perform clustering on stored session data to identify recurring behavioral patterns among users. This allows segmentation of audiences into distinct behavioral groups, such as highly engaged users, passersby, or intermittent viewers.

Such insights are valuable for understanding customer interaction styles and refining advertising strategies.

1.1.2 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance constraints, and operational expectations of the system. These requirements ensure that the system is not only functionally correct but also efficient, reliable, and suitable for real-world deployment.

Real-Time Performance

The system must process incoming video frames and generate engagement outputs with minimal latency. Real-time responsiveness is critical for enabling dynamic advertisement triggering and maintaining system effectiveness.

Efficient algorithms and optimized pipelines are required to ensure smooth and continuous operation without delays.

Scalability

The system should be designed in a modular and extensible manner, allowing it to scale with increasing data volume, additional camera inputs, or deployment across multiple mall locations. This ensures long-term usability and adaptability to growing commercial requirements.

Robustness

The system must operate reliably under diverse environmental conditions, including varying lighting, camera angles, crowd densities, and background noise. It should maintain stable performance even in the presence of occlusions, fast movements, and unpredictable real-world dynamics.

Accuracy

Engagement classification must achieve acceptable levels of precision and recall to ensure meaningful and trustworthy outputs. High false positive rates can lead to incorrect advertisement triggers, while false negatives may result in missed engagement opportunities. Therefore, maintaining a balance between sensitivity and specificity is essential.

Computational Efficiency

Given real-world deployment constraints, the system must utilize lightweight models and optimized processing techniques to minimize computational overhead. This ensures that the system can operate effectively on moderate hardware without requiring specialized GPU infrastructure, making it more accessible and cost-effective.

Maintainability and Modularity

The system should follow a modular design approach, where individual components such as detection, tracking, and analytics modules can be independently updated or replaced. This facilitates easier debugging, system upgrades, and long-term maintenance.

Cost-Effectiveness

Cost-efficiency is a critical requirement, particularly in the context of Sri Lankan retail environments. The system should operate using affordable hardware and avoid reliance on expensive infrastructure or proprietary solutions. This ensures practical feasibility and supports wider adoption across different mall settings.

Reliability and Continuous Operation

The system must be capable of continuous operation over extended periods without failure. It should handle runtime exceptions gracefully and ensure data consistency during long-term usage. Reliable performance is essential for real-world deployment in commercial environments where downtime can impact business operations.

2.2 Feasibility Study

A feasibility study was conducted to evaluate the practicality, viability, and sustainability of the proposed real-time context-aware engagement analytics system. This analysis ensures that the system can be successfully developed and deployed within the constraints of Sri Lankan mall environments while meeting both technical and commercial expectations. The feasibility assessment is carried out across multiple dimensions, including technical, economic, operational, and legal considerations.

2.2.1 Technical Feasibility

Technical feasibility evaluates whether the system can be implemented using existing technologies and infrastructure. The proposed solution is built on established computer vision techniques such as real-time object detection, multi-object tracking, and behavioral analytics.

The system is optimized for efficiency using lightweight models and streamlined processing pipelines, enabling real-time performance on moderate hardware without requiring high-end GPUs. This makes it suitable for cost-sensitive mall environments.

It also integrates standard tools for data storage and analytics, such as NoSQL databases and machine learning libraries for clustering and forecasting. Overall, the system is technically feasible using current industry-standard technologies.

2.2.2 Economic Feasibility

Economic feasibility assesses the cost-effectiveness of developing and deploying the system. The proposed solution is designed to minimize costs by utilizing existing surveillance cameras and open-source software, reducing the need for specialized hardware and expensive licensing.

Operational costs are also kept low through lightweight models that require less computing power, resulting in lower energy consumption and maintenance expenses. From a business perspective, the system adds value by enabling data-driven advertising, improving customer engagement, and increasing potential return on investment.

Overall, the benefits of improved advertising efficiency and behavioral insights outweigh the implementation and operational costs, making the system economically viable.

2.2.3 Operational Feasibility

Operational feasibility evaluates how effectively the system can function in real-world environments and integrate with existing workflows. The proposed system is tailored for dynamic mall settings, addressing challenges such as crowd density, lighting variations, and movement patterns through context-aware mechanisms like spatial zone analysis and adaptive thresholds.

It is designed for continuous, automated operation with minimal human intervention, supported by real-time analytics and structured data logging. Additionally, the system can be easily integrated

with existing digital signage infrastructure, enabling smooth deployment without major operational changes.

Its modular and scalable architecture further supports easy updates and expansion across multiple locations, ensuring practical long-term usability.

2.2.4 Legal and Ethical Feasibility

Legal and ethical feasibility focuses on privacy, consent, and responsible system use. Since the system processes video data, it is designed to prioritize privacy by focusing on behavioral analysis rather than personal identification, without using facial recognition or storing personally identifiable information.

Data is anonymized and stored in aggregated form to support privacy protection and compliance with data handling principles. Informational signage can also be used in mall environments to ensure transparency and user awareness.

Ethically, the system aims to improve user experience and advertising effectiveness rather than monitor individuals intrusively. This ensures responsible use of data and supports public acceptance of the system.

2.2.5 Schedule Feasibility

Schedule feasibility evaluates whether the system can be developed within the available time frame.

The development process is divided into clearly defined phases, including requirement analysis, system design, model development, integration, testing, and evaluation. Each phase can be completed incrementally, allowing continuous progress and validation.

Given the use of existing tools, frameworks, and pre-trained models, the development timeline is realistic and achievable within an academic project duration. Proper planning and modular implementation further support timely completion.

Based on the analysis, the proposed real-time engagement analytics system is feasible from technical, economic, operational, and ethical perspectives. The system leverages existing technologies, minimizes costs, aligns with real-world deployment conditions, and adheres to responsible data practices.

Therefore, the feasibility study confirms that the proposed solution can be successfully implemented and deployed in Sri Lankan mall environments, providing both practical value and research contribution.

2.3 System Diagram

The proposed system adopts a layered, dual-output architecture designed to support real-time engagement detection while simultaneously enabling long-term behavioral analytics. This architectural design separates low-level perception tasks from higher-level decision-making and analytical processes, ensuring both computational efficiency and system scalability.

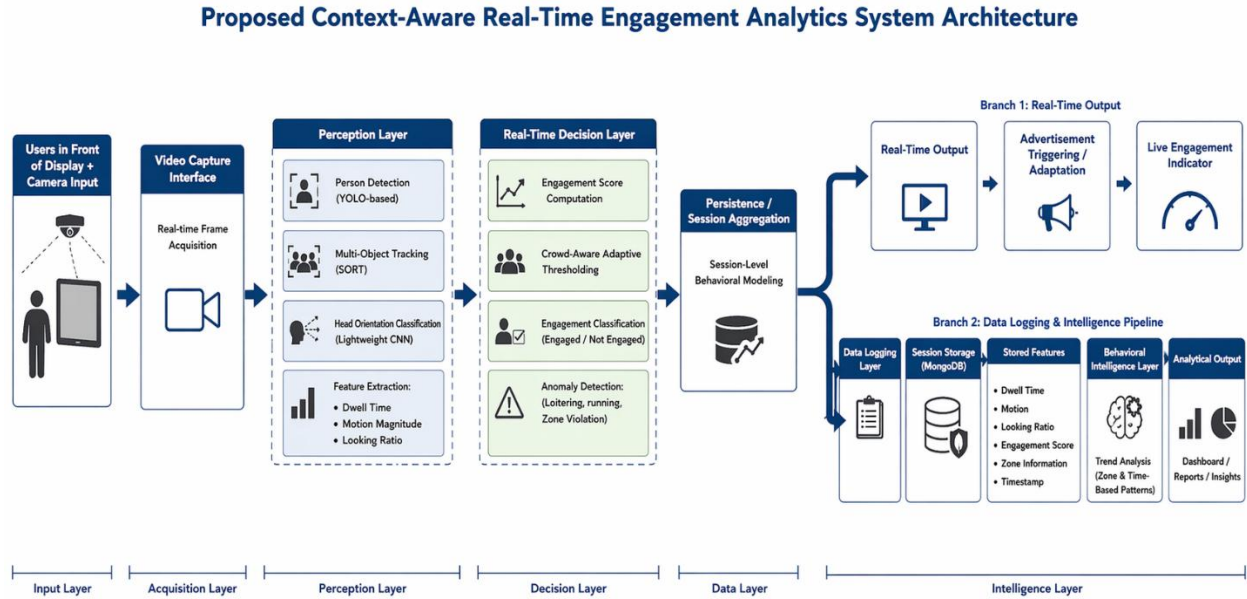


Figure 2: Overall System Architecture Diagram

The architecture is structured into four primary layers: the Perception Layer, Real-Time Decision Layer, Data Logging Layer, and Behavioral Intelligence Layer. These layers operate sequentially while maintaining modular independence, allowing each component to be optimized or updated without affecting the overall system functionality.

The system processes continuous video input from a camera positioned in front of an advertising display. The captured frames are passed through the perception pipeline, transformed into behavioral signals, and then used to generate both real-time outputs and persistent analytical data.

2.3.1 Perception Layer

The perception layer extracts low-level visual and behavioral information from the video stream in real time. It serves as the foundation of the system by continuously monitoring the environment.

It uses a YOLO-based person detection module to identify individuals in each frame, producing bounding boxes for spatial localization. These detections are then passed to the SORT tracking algorithm, which assigns consistent identities across frames to enable trajectory tracking and temporal feature extraction.

In parallel, a CNN-based head orientation model analyzes cropped person images to estimate visual attention toward the display. The system also computes motion and dwell-time features from tracking data, capturing movement patterns and time spent in the monitored area.

Together, these components convert raw video input into structured behavioral signals for higher-level engagement analysis.

2.3.2 Real-Time Decision Layer

The real-time decision layer processes behavioral signals to assess engagement and detect anomalies, operating continuously to support immediate system responses.

It first computes an engagement score by combining features such as looking ratio, dwell time, and motion magnitude. This score represents the likelihood of a user interacting with the display. Zone-aware refinement is then applied by adjusting scores based on spatial regions like interaction zones and walkways, enabling context-sensitive interpretation.

Adaptive thresholding further improves reliability by dynamically adjusting decision limits according to crowd density and historical patterns. Based on the refined score, the system classifies users as engaged or not engaged as the primary real-time output.

Additionally, an anomaly detection module identifies unusual behaviors such as abnormal dwell times or motion patterns, improving system robustness and consistency.

2.3.3 Data Logging Layer

The data logging layer is responsible for storing structured behavioral data for long-term analysis. This layer ensures that valuable information generated during real-time processing is not lost and can be used for further analytics.

The system uses MongoDB, a NoSQL database, to store session-level data. Each interaction session is recorded as a structured document containing attributes such as dwell time, looking ratio, motion statistics, zone information, timestamps, and engagement scores.

By maintaining a persistent record of user interactions, the system enables longitudinal analysis and supports the development of advanced behavioral intelligence features. The use of MongoDB provides flexibility in handling semi-structured data and supports scalability as the volume of stored data increases.

2.3.4 Behavioral Intelligence Layer

The behavioral intelligence layer analyzes stored session data to generate higher-level insights and predictions. Unlike the real-time layer, it operates offline or semi-online, enabling more complex analysis without affecting system performance. This supports audience segmentation and targeted advertising.

The layer also conducts engagement trend analysis by aggregating data over time and space to identify patterns such as peak hours and high-performing zones, helping optimize advertisement scheduling.

2.3.5 Dual-Output System Design

A key feature of the proposed architecture is its dual-output design, which enables the system to generate two types of output simultaneously:

1. Real-Time Output:
Immediate engagement classification results used to trigger or adapt advertisements dynamically.
2. Analytical Output:
Session-level data stored for long-term analysis, clustering, and forecasting.

This dual-output approach ensures that the system provides both operational functionality and strategic intelligence. While the real-time output supports immediate decision-making, the analytical output enables continuous learning and system improvement.

By combining real-time processing with data-driven intelligence, the system overcomes the limitations of traditional advertising systems and enables context-aware engagement analytics suitable for real-world deployment. The modular design ensures flexibility, scalability, and maintainability, making the system adaptable to different environments and future technological advancements.

2.4 Data Collection and Preprocessing

2.4.1 Data Collection and Sources

Data collection for the proposed system was conducted using a hybrid approach that combines primary data acquisition and publicly available datasets to ensure both realism and model generalization.

Primary Data Collection

Primary data was collected using a camera positioned in front of a simulated advertising display environment. The setup was designed to closely resemble real-world mall scenarios, allowing the system to capture natural human behaviors in an interactive context. The camera continuously recorded video streams covering a variety of situations, including:

- Individual passers-by moving through the scene
- Stationary viewers interacting with the display
- Group and crowd movement patterns
- Variations in environmental conditions such as lighting intensity and shadows

This setup enabled the collection of realistic behavioral data required for testing and validating the engagement analytics pipeline. The collected dataset includes multiple levels of information generated during processing.

Use of Publicly Available Dataset

In addition to primary data, the study also utilized the publicly available Mall Dataset to support model training and validation. The Mall Dataset contains annotated video sequences of crowded shopping mall environments, capturing realistic pedestrian behaviors, density variations, and movement patterns.

The inclusion of this dataset provides several advantages:

- Exposure to diverse crowd scenarios and environmental conditions
- Improved robustness of detection and tracking models
- Enhanced generalization capability across different mall settings

The Mall Dataset was primarily used during the training and evaluation phases of detection and behavioral modeling components, ensuring that the system performs effectively beyond the controlled simulation environment.

2.4.2 Data Preprocessing

Data preprocessing is a critical stage that transforms raw video data into structured and consistent inputs suitable for modeling and analysis. Given the variability of real-world environments, preprocessing ensures that the extracted features are reliable, normalized, and robust to noise.

The key preprocessing steps include:

Frame resizing and normalization:

All video frames are resized to a uniform resolution to ensure consistent processing and reduce computational overhead. Pixel values are normalized to improve model performance and stability.

Noise reduction:

Basic filtering techniques are applied to reduce visual noise caused by lighting variations, shadows, and camera artifacts, thereby improving detection accuracy.

Bounding box extraction and refinement:

Detected regions corresponding to individuals are extracted and refined to ensure accurate spatial localization and minimize detection errors.

Feature scaling:

Behavioral features such as dwell time and motion magnitude are scaled to a consistent range. This prevents features with larger numerical values from dominating the engagement computation process.

Temporal smoothing of attention signals:

Head orientation predictions are smoothed across consecutive frames to reduce fluctuations caused by rapid or minor head movements. This improves the stability and reliability of attention estimation.

Preprocessing plays a vital role in ensuring that the input data is clean, consistent, and suitable for both real-time inference and long-term behavioral analysis. By combining data from simulated environments and publicly available datasets, the system achieves a balance between controlled experimentation and real-world generalization.

2.5 Model Design

The proposed system combines multiple computer vision and machine learning models for real-time behavioral analysis and engagement estimation using modular architecture. This design improves flexibility, maintainability, and scalability by allowing individual components to be optimized independently.

Its core components include a YOLO-based person detection model, SORT tracking, and a lightweight CNN for head orientation classification. These work sequentially to convert raw visual data into structured behavioral insights.

The system is designed to bridge perception and behavioral intelligence by transforming spatial and temporal tracking data into meaningful engagement features for higher-level analysis. The design prioritizes:

- Real-time performance (low latency processing)
- Computational efficiency (lightweight models)
- Robustness (handling noise, occlusion, and crowd dynamics)
- Scalability (support for multi-camera deployment)

2.5.1 Person Detection Model

The first stage of the perception pipeline involves detecting individuals in each video frame using the YOLO object detection model. YOLO is selected due to its ability to perform object detection in a single forward pass, making it highly suitable for real-time applications.

Given an input frame I_t , the YOLO model produces a set of bounding boxes:

$$B_t = \{b_t^1, b_t^2, \dots, b_t^{N_t}\}$$

where N_t represents the number of detected individuals at time t . Each bounding box b_t^i contains spatial attributes such as position, width, height, and confidence score. The detection module performs the following key functions:

- Identifies all visible individuals within the scene
- Provides accurate spatial localization for each detected person
- Serves as the foundation for tracking and behavioral analysis

To enhance robustness, the detection model is configured to handle:

- Variations in lighting conditions
- Background clutter and occlusions
- Different crowd densities

The efficiency of YOLO ensures that detection can be performed in real time while maintaining a balance between speed and accuracy.

2.5.2 Multi-Object Tracking (SORT)

Following detection, the system employs the SORT (Simple Online and Realtime Tracking) algorithm to maintain persistent identities for individuals across frames. SORT combines Kalman filtering for motion prediction and Hungarian algorithm-based data association for matching detections.

For each individual, SORT generates a trajectory:

$$\{p_i(1), p_i(2), \dots, p_i(T_i)\}$$

where $p_i(t)$ represents the centroid position of individual i at frame t , and T_i is the total duration of the tracked session. The tracking module enables:

- Identity consistency across frames
- Temporal continuity for behavior analysis
- Trajectory generation for motion modeling

This component is essential for transitioning from isolated frame-level detections to continuous session-based behavioral understanding.

2.5.3 Head Orientation Model (Lightweight CNN)

To estimate user attention, the system incorporates a custom-designed lightweight CNN that classifies whether an individual is looking toward the display.

CNN Architecture

The architecture is optimized for efficiency while maintaining sufficient accuracy. It includes:

- Convolutional layers for spatial feature extraction
- ReLU activation functions for non-linearity
- Max pooling layers for dimensionality reduction
- Fully connected layers for classification
- Sigmoid output layer for binary prediction
- The model processes cropped head regions and outputs:

$$A_i(t) \in \{0,1\}$$

- $A_i(t) = 1$: individual is looking at the display
- $A_i(t) = 0$: individual is not looking

This binary attention signal is later aggregated over time to produce stable attention metrics.

2.5.4 Behavioral Feature Computation

Based on the outputs of the detection, tracking, and head orientation models, the system computes three primary behavioral features:

1. Looking Ratio

The looking ratio represents the proportion of time an individual spends looking at the display:

$$L_i = \frac{1}{T_i} \sum_{t=1}^{T_i} A_i(t)$$

These metric captures sustained attention rather than momentary glances.

2. Dwell Time

Dwell time is defined as the total duration for which an individual remains within the monitored area:

$$D_i = T_i \cdot \Delta t$$

where Δt is the time interval between frames.

Dwell time reflects the level of persistence and potential interest in the display.

3. Motion Magnitude

Motion magnitude measures the level of movement exhibited by an individual:

$$M_i = \frac{1}{T_i - 1} \sum_{t=2}^{T_i} \| p_i(t) - p_i(t-1) \|$$

High motion values indicate transient behavior, while lower values suggest stationary or engaged behavior.

2.5.5 Engagement Score Calculation

The engagement model integrates the three behavioral features into a single quantitative measure. The engagement score is defined as:

$$E_i = w_1 L_i + w_2 \tilde{D}_i - w_3 \tilde{M}_i$$

- L_i = looking ratio
- $\tilde{D}_i$ = normalized dwell time
- $\tilde{M}_i$ = normalized motion magnitude
- w_1, w_2, w_3 = weighting coefficients

The engagement score combines:

- Attention (Looking Ratio) – positive contribution
- Persistence (Dwell Time) – positive contribution
- Movement (Motion) – negative contribution

This formulation ensures that individuals who remain in the area exhibit consistent attention, and show minimal movement are classified as highly engaged.

2.5.6 Engagement Classification

Based on the computed engagement score, individuals are classified as engaged or not engaged using a threshold:

$$\hat{y}_i = \begin{cases} 1 & \text{if } E_i \geq T_{engage} \\ 0 & \text{otherwise} \end{cases}$$

where T_{engage} is a predefined or adaptive threshold.

This classification step serves as the final decision layer in the real-time inference pipeline. It translates continuous engagement scores into discrete outputs that can be directly utilized by downstream components, particularly for triggering advertisement responses.

The threshold T_{engage} can be configured in two ways:

- **Fixed Threshold:** A constant value determined empirically during system calibration, ensuring consistent behavior across different environments.
- **Adaptive Threshold:** A dynamic value that adjusts based on contextual factors such as crowd density, spatial zones, or historical engagement patterns, thereby improving robustness in varying conditions.

To enhance reliability, classification is often applied on temporally smoothed engagement scores rather than raw instantaneous values.

This reduces sensitivity to noise, momentary attention shifts, and tracking inconsistencies, resulting in more stable and consistent predictions.

Overall, the engagement classification module plays a critical role in bridging behavioral analysis and actionable system response, ensuring that only meaningful interactions influence advertisement decisions.

2.5.7 Contextual Refinement of Engagement

While the initial engagement score provides a quantitative estimate based on behavioral features, real-world environments introduce variability that cannot be fully captured by feature-level modeling alone. To address this limitation, the proposed system incorporates a contextual refinement layer that adjusts engagement estimates using spatial, environmental, and temporal context. This layer enhances robustness, reduces false positives, and improves the interpretability of engagement predictions.

1. Zone-Based Weighting

Engagement behavior is highly dependent on the spatial location of an individual within the scene. On account for this, the environment is divided into semantic zones, such as:

- Interaction zone (directly in front of the display)
- Peripheral zone (near the display but not directly interacting)
- Walkway or transit zone (areas with continuous movement)
- Entry/exit zones

Each zone is assigned a weighting factor that reflects its relevance to engagement. Individuals located in interaction zones are more likely to be genuinely engaged, while those in walkways are more likely to be transient.

This ensures that identical behavioral patterns are interpreted differently depending on spatial context, thereby improving classification accuracy.

2. Crowd-Aware Thresholding

Crowd density significantly influences human behavior and system observations. In high-density environments, individuals may exhibit reduced movement or longer dwell times due to congestion rather than genuine engagement. Conversely, in low-density settings, shorter dwell times may still indicate interest.

To address this, the system dynamically adjusts the engagement threshold based on crowd density.

This adaptive thresholding mechanism ensures that:

- Engagement is not overestimated in crowded conditions
- Engagement is not underestimated in sparse environments

By incorporating crowd awareness, the system maintains consistent performance across varying environmental conditions.

3. Temporal Smoothing

Real-time predictions are inherently subject to fluctuations due to noise, momentary gaze shifts, or tracking inconsistencies. To reduce instability, the system applies temporal smoothing to engagement scores or classification outputs.

This process:

- Reduces abrupt changes in engagement classification
- Improves temporal consistency
- Provides more stable output for real-time decision-making

Impact on System Performance

The integration of contextual refinement provides several advantages:

- Improved accuracy by reducing false positives in crowded or irrelevant zones
- Enhanced robustness under real-world variability such as occlusion and noise
- Better interpretability of engagement decisions through context-aware reasoning
- Stability in real-time outputs, enabling reliable advertising triggering

The proposed model integrates detection, tracking, and classification into a unified engagement analytics framework using YOLO, SORT, and a lightweight CNN. It efficiently extracts behavioral features to generate real-time engagement insights.

Through session-based modeling and contextual refinement, the system ensures robust performance and low computational cost. This makes it suitable for accurate real-time analysis as well as long-term behavioral intelligence in dynamic mall environments.

2.5 Commercialization Aspects of the Product

The proposed real-time engagement analytics system is designed not only as a research prototype but also as a commercially viable solution for intelligent advertising in modern retail environments. By combining real-time behavioral analytics with long-term data-driven insights, the system provides tangible value for businesses seeking to optimize customer engagement and advertising effectiveness.

2.5.1 Market Relevance

The system addresses a growing demand for data-driven advertising and smart retail solutions. It is highly suitable for deployment in environments where large volumes of people interact with digital displays, including:

- Shopping malls
- Airports and transportation hubs
- Retail stores and showrooms
- Exhibition centers and public venues

Traditional digital signage systems operate in a static manner, delivering the same content regardless of audience behavior. In contrast, the proposed system enables dynamic and targeted advertising, where content can be adapted based on real-time audience engagement.

This capability allows businesses to:

- Deliver more relevant advertisements
- Increase audience attention and interaction
- Improve overall customer experience

As a result, the system contributes directly to improving advertising efficiency and return on investment (ROI).

2.5.2 Business Value

The system generates significant business value by transforming raw visual data into actionable insights. Key benefits include:

Real-time audience engagement insights

Businesses can monitor how customers interact with displays, enabling immediate adjustments to advertising strategies. This real-time feedback helps identify which content attracts the most attention, how long users engage with ads, and which demographics respond most effectively.

Behavioral segmentation

By analyzing patterns such as dwell time, attention, and movement, the system can categorize audiences into different behavioral groups (e.g., engaged viewers, passers-by, casual observers). This supports targeted marketing strategies.

Predictive analytics for advertising scheduling

Using historical data and forecasting models, the system can identify peak engagement periods and recommend optimal times for displaying specific advertisements.

Performance measurement

Advertisers can evaluate the effectiveness of campaigns based on actual audience engagement rather than assumptions, leading to more informed decision-making.

Overall, the system shifts advertising from a passive medium to an intelligent, feedback-driven process.

2.5.3 Deployment Considerations

A key strength of the proposed system is its practical deployability, especially in emerging retail markets such as Sri Lanka.

- **Low-cost hardware compatibility**
The system can operate using standard surveillance cameras, eliminating the need for expensive specialized equipment.
- **CPU-based operation**
By utilizing lightweight models and optimized pipelines, the system can run on CPU-based systems without requiring high-end GPUs. This significantly reduces infrastructure costs.
- **Scalability**
The modular architecture allows the system to be deployed across multiple locations, such as different stores within a mall or across multiple malls. Centralized data management can further support large-scale analytics.
- **Ease of integration**
The system can be integrated with existing digital signage platforms, minimizing disruption to current operations.
- **Robustness to real-world conditions**
The system is designed to handle variations in lighting, crowd density, and environmental conditions, ensuring reliable performance in practical settings.

These considerations ensure that the system is not only technically effective but also commercially deployable and sustainable.

2.5.4 Revenue Opportunities

The proposed system opens multiple avenues for revenue generation and business expansion:

- **Advertisement optimization services**
Businesses can use engagement analytics to refine and optimize advertisement strategies, leading to increased advertising value.
- **Data analytics services**
The system can provide detailed reports and insights to retailers and advertisers, which can be offered as a subscription-based service.
- **Smart retail integration**
Integration with broader smart retail ecosystems (e.g., recommendation systems, customer analytics platforms) can enhance overall business intelligence.
- **Software-as-a-Service (SaaS) model**
The system can be deployed as a cloud-based or subscription-based solution, allowing multiple clients to access analytics without maintaining their own infrastructure.
- **Custom analytics solutions**
Businesses may request tailored analytics modules, creating additional opportunities for customization and service-based revenue.

2.5.5 Competitive Advantage

The proposed system offers a significant improvement over traditional digital advertising by introducing context-aware engagement modeling instead of relying on simple signals like presence or gaze. It integrates spatial zones, crowd density, and temporal factors, resulting in more accurate and meaningful engagement detection.

It also features a dual-output architecture that provides both real-time analytics and long-term behavioral insights. This enables immediate response to user behavior while also supporting historical analysis and predictive decision-making.

The system is designed to be cost-effective and deployable on CPU-based hardware, making it suitable for environments with limited resources. In addition, it enables advanced capabilities such as audience segmentation, anomaly detection, and engagement forecasting through behavioral intelligence.

2.5.6 Challenges and Considerations

Despite strong commercialization potential, the proposed system faces several practical challenges for real-world deployment.

A major challenge is environmental variability, including changes in lighting, camera angles, and crowd density. These factors can introduce occlusions and noise, affecting detection, tracking, and engagement estimation accuracy.

Computational efficiency is another concern, as real-time processing of continuous video streams requires optimized resource usage, even with lightweight models. Hardware limitations may still impact performance in large-scale deployments. Data privacy is also critical, since the system operates in public spaces. It must ensure privacy by avoiding storage of personally identifiable information and focusing only on behavioral analytics.

In addition, system calibration may vary across locations, requiring adjustments to thresholds, zone definitions, and sensitivity based on each mall's layout and customer behavior patterns.

Finally, reliable data storage and network stability are important for continuous analytics, requiring scalable storage solutions such as cloud or hybrid systems to maintain data integrity over time.

2.5.7 Future Commercial Enhancements

The proposed system can be further enhanced to increase its commercial value and scalability. One key improvement is AI-driven personalized advertising, where engagement data is combined with behavioral or demographic insights to deliver targeted content.

A cloud-based analytics platform can also be introduced to enable centralized monitoring across multiple locations and support large-scale trend analysis. In addition, real-time adaptive advertising can be implemented, allowing content to change dynamically based on live engagement levels.

Future work may also include integration with external systems such as point-of-sale, mobile apps, and loyalty programs for a more complete view of customer behavior. Advanced deep learning and transformer models could further improve prediction accuracy and engagement analysis.

Ultimately, the system could evolve into a comprehensive smart retail analytics platform supporting advertising optimization, customer flow analysis, and store planning.

2.6 Testing & Implementation

2.6.1 Implementation

Tools and Technologies

The proposed system was implemented using a carefully selected set of technologies that support real-time processing, scalability, and efficient integration of computer vision and machine learning components. Each technology plays a specific role within the system architecture, contributing to the overall performance and functionality.

Programming Language

- Python was used as the primary development language due to its simplicity, flexibility, and extensive ecosystem of libraries for machine learning and computer vision.
- It enables rapid prototyping, seamless integration of models, and efficient handling of data processing pipelines.

Computer Vision Library

OpenCV (Open-Source Computer Vision Library) is used for handling image and video processing tasks. OpenCV provides functionalities such as:

- Video frame capture
- Image manipulation
- Face detection support
- Real-time processing

Its efficiency and real-time capabilities make it suitable for handling continuous video streams in the system.

Machine Learning Frameworks

- TensorFlow and Keras were utilized for developing, training, and deploying machine learning models.
- Keras, as a high-level API, simplifies model design, while TensorFlow provides efficient backend computation and scalability.
- These frameworks support real-time inference and are well-suited for deploying lightweight neural networks.

Deep Learning Models

The system integrates multiple deep learning models to perform different tasks within the pipeline. A YOLO (You Only Look Once) model is used for real-time person detection due to its ability to process entire images in a single forward pass, making it highly efficient for live applications. For tracking detected individuals across frames, the SORT (Simple Online and Realtime Tracking) algorithm is employed. This tracking method ensures consistent identity assignment and enables the computation of temporal behavioral features such as dwell time and movement patterns. Additionally, a custom lightweight convolutional neural network is used to classify head orientation, providing a reliable estimation of whether an individual is looking at the display.

Backend Framework

The backend of the system is implemented using Python-based processing pipelines that handle the core functionality of the application. This includes capturing and processing video streams, executing detection and tracking algorithms, performing behavioral feature extraction, computing

engagement scores, and managing communication with the database. The backend is designed in a modular manner, ensuring that each functional component can operate independently while contributing to the overall workflow.

Frontend Framework

The user interface of the system is developed using React, a widely used JavaScript library for building dynamic and interactive web applications. React provides a component-based architecture that allows efficient management of user interface elements and real-time updates.

In the system, React is used to display advertisements dynamically based on system outputs. It enables smooth rendering of visual content and provides interactive user experience. The framework also supports efficient state management, ensuring that the interface responds quickly to changes in user data and system predictions.

Database System

For data storage, MongoDB is used as the primary database system. MongoDB is a NoSQL database that stores data in a flexible, JSON-like format, making it well-suited for handling dynamic and semi-structured behavioral data. The database stores session-level information, including tracking data, behavioral features such as dwell time and motion, and computed engagement scores. This structured storage enables long-term analytics, clustering, and forecasting, which are essential for deriving meaningful business insights.

Hardware Components

The system operates using standard hardware components, including:

- Webcam or camera for video input
- Processing unit (CPU/GPU) for model inference
- Display screen for advertisement output

In some cases, GPU acceleration may be used to improve performance in real-time processing.

Deployment Environment

The deployment environment is designed to be flexible and scalable. The system can run on both Windows and Linux operating systems and can process live video streams in real time.

Deployment options include:

- Edge deployment: Processing occurs locally on-site for low latency
- Hybrid/cloud deployment: Data can be stored and analyzed remotely for large-scale analytics

Additionally, the system can be scaled across multiple locations, allowing centralized monitoring and management in commercial retail settings.

Table 2: Tools and Technologies Used in the System

Component	Technology
Programming Language	Python
Machine Learning Framework	TensorFlow / Keras
Deep Learning Model (Detection)	YOLO
Multi-Object Tracking	SORT
Head Orientation Model	Lightweight CNN
Computer Vision Library	OpenCV
Backend Processing	Python Pipeline (Modular Scripts)
Frontend Interface	React
Database System	MongoDB
Data Format	JSON (Document-Based Storage)
Operating System	Windows / Linux
Hardware	CPU-based System + Camera
Deployment	Edge-based / Local Deployment

2.6.2 Testing and Evaluation

To ensure the reliability and effectiveness of the proposed system, a comprehensive testing strategy was implemented. The system was evaluated under multiple scenarios to simulate real-world conditions and assess its performance across different behavioral patterns.

Unit Testing

Unit testing was conducted to verify the correctness and functionality of individual system components independently. Each module of the system was tested in isolation to ensure that it performs as expected before integration.

The key components tested include:

- YOLO Person Detection Module
Verified the accuracy of detecting individuals in different lighting conditions and backgrounds.
- SORT Tracking Module
Tested for correct assignment of unique IDs and consistent tracking across frames.
- CNN Head Orientation Model
Evaluated classification accuracy for detecting whether a person is looking at the display.
- Feature Extraction Module
Verified correctness of computed behavioral features such as dwell time, motion magnitude, and looking ratio.
- Engagement Scoring Module
Tested whether engagement scores are correctly calculated based on input features.

Unit testing ensured that each component produces valid outputs, reducing the likelihood of errors during system integration.

The integration of behavioral feature extraction with engagement scoring was tested to confirm that computed features are correctly processed to generate engagement scores. This involved validating feature consistency, normalization processes, and the correctness of score computation under different behavioral conditions.

Additionally, the integration of engagement outputs with MongoDB logging was tested to ensure that session-level data is stored accurately and efficiently. This included verifying data structure consistency, successful insertion of records, and the ability to retrieve stored data for further analysis.

The primary objectives of integration testing were to:

- Ensure correct and efficient data flow between system components
- Verify compatibility of input and output formats across modules
- Maintain synchronization across real-time processing stages
- Confirm consistent and stable operation of the complete pipeline

This phase confirmed that the system functions cohesively as a unified framework, with all components working together to deliver accurate real-time engagement analytics.

This phase confirmed that the system operates cohesively as a unified framework.

System Testing

System testing was conducted to evaluate the complete system as an integrated solution under real-world-like conditions. Unlike unit and integration testing, it focused on end-to-end functionality—from video input to engagement output and data storage.

The system was tested using continuous video streams to simulate real operational conditions. These tests ensured that all modules operate synchronously and produce coherent results when functioning together.

Key evaluations included multi-person detection and tracking in both low- and high-density scenarios, accuracy of engagement classification, and anomaly detection for unusual behaviors.

Finally, the reliability of session-level data storage in MongoDB was verified. This included ensuring that all relevant behavioral data, engagement scores, and session information were correctly recorded and retrievable for further analysis.

Overall, system testing confirmed that the proposed solution operates effectively as a unified framework, producing accurate, consistent, and meaningful outputs under realistic conditions.

Real-Time Performance Evaluation

The system was also evaluated for real-time performance, which is essential for timely engagement detection and advertisement triggering.

Key metrics included latency, which measures frame processing delay, frame rate (FPS), which indicates processing speed per second, and CPU utilization, which reflects system resource efficiency. These metrics ensure the system operates smoothly without excessive delays or hardware load.

Testing was conducted under varying crowd densities and lighting conditions, confirming that the system maintains stable real-time performance and is suitable for deployment in cost-sensitive environments.

Evaluation Metrics

The performance of the engagement classification model was evaluated using standard classification metrics to ensure a comprehensive assessment of accuracy, reliability, and robustness.

Precision measures the proportion of correctly predicted engaged instances out of all predicted engaged cases:

$$\text{Precision} = \frac{TP}{TP + FP}$$

This metric reflects how accurate the system is in identifying true engagement without generating false positives.

Recall evaluates the system’s ability to correctly identify all actual engaged instances:

$$\text{Recall} = \frac{TP}{TP + FN}$$

A high recall indicates that the system successfully captures most of the true engagement cases.

F1-Score provides a balanced measure by combining precision and recall:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

This metric is particularly useful when there is an imbalance between engaged and non-engaged instances.

False Positive Rate (FPR) measures the proportion of non-engaged individuals incorrectly classified as engaged:

$$\text{FPR} = \frac{FP}{FP + TN}$$

A lower FPR is important to prevent incorrect advertisement triggering and ensure system reliability.

Where:

- TP (True Positives): Correctly identified engaged individuals
- FP (False Positives): Non-engaged individuals incorrectly classified as engaged
- FN (False Negatives): Engaged individuals incorrectly classified as non-engaged
- TN (True Negatives): Correctly identified non-engaged individuals

These metrics collectively provide a comprehensive evaluation of the classification model, ensuring that both accuracy and reliability are considered.

Test Case Design

Test cases were designed to represent typical user behaviors observed in mall environments. These cases were used to evaluate whether the system can correctly distinguish between different engagement levels.

Table 3: Test Case for High Engagement Detection

Test Case ID	TC-01
Test Case	High Engagement Detection
Scenario	A person stands in front of the display and continuously looks at it
Test Input Data	Video frames with a stationary individual facing the display

Test Procedure	Capture frames → YOLO detection → SORT tracking → CNN head orientation → Feature computation → Engagement scoring
Expected Outcome	High engagement score and classified as engaged
Actual Outcome	Engagement score exceeded threshold and classified as engaged
Test Result	Pass

Table 4: Test case for Low Engagement Detection

Test Case ID	TC-02
Test Case	Low Engagement Detection
Scenario	A person walks past the display without stopping
Test Input Data	Video frames showing fast-moving individual
Test Procedure	YOLO detection → SORT tracking → Head orientation → Feature calculation → Engagement scoring
Expected Outcome	Low engagement score and classified as not engaged
Actual Outcome	Engagement score below threshold and classified as not engaged
Test Result	Pass

Table 5: Test Case for Loitering Detection

Field	Description
Test Case ID	TC-03
Test Case	Loitering Detection
Scenario	People remain in area for long duration
Test Input Data	Prolonged presence video frames
Test Procedure	Track → Compute dwell time → Compare threshold
Expected Outcome	System flags anomaly
Actual Outcome	Loitering detected
Test Result	Pass

Table 6: test Case for Running

Test Case ID	TC-04
Test Case	Erratic Motion Detection
Scenario	Person moves irregularly
Test Input Data	Rapid motion video frames
Test Procedure	Track → Compute motion → Compare threshold
Expected Outcome	Abnormal motion detected
Actual Outcome	Motion threshold exceeded
Test Result	Pass

RESULTS AND DISCUSSION

3.1 Results

This section presents the experimental results obtained from the implementation and evaluation of the proposed real-time engagement analytics system. The performance of the system is analyzed in terms of detection accuracy, engagement classification, real-time processing capability, and behavioral intelligence outputs.

The system integrates two primary models:

- A YOLO-based person detection model for identifying individuals in video frames
- A lightweight CNN-based head orientation model for estimating visual attention

These models collectively enable the extraction of meaningful behavioral signals such as dwell time, motion patterns, and attention ratio, which are then used to compute engagement scores and classify user behavior.

Overall, the integration of these components allows the system to transform raw video input into structured behavioral intelligence suitable for real-time advertising analytics.

3.1.1 Person Detection Model Results

The YOLO-based person detection model was evaluated to determine its effectiveness in identifying individuals within real-time video streams under varying environmental and operational conditions. The evaluation was conducted using simulated mall scenarios, including variations in lighting conditions, crowd densities, and partial occlusions. The model achieved:

$$\text{mAP}@0.5 = 0.89$$

$$\text{Precision} = 0.87$$

$$\text{Recall} = 0.37$$

The precision and recall values were observed, indicating a good balance between correct detections and missed detections.

In well-illuminated environments, the model consistently produced high-confidence detections with minimal false positives. In moderately crowded scenarios, the system maintained stable performance, successfully detecting multiple overlapping individuals with only minor reductions in confidence scores.

Under heavy occlusion and high-density crowd conditions, a slight decrease in recall was observed due to overlapping bounding regions; however, the system continued to produce acceptable detection outputs suitable for downstream processing.

From a real-time performance perspective, the model achieved an average processing speed of 25–35 FPS, with an inference latency of approximately 30–45 milliseconds per frame, making it suitable for real-time deployment in mall environments.

Overall, the YOLO-based detection model provides a robust, efficient, and real-time capable perception layer, ensuring reliable identification of individuals in dynamic environments, which is critical for the success of the proposed system.

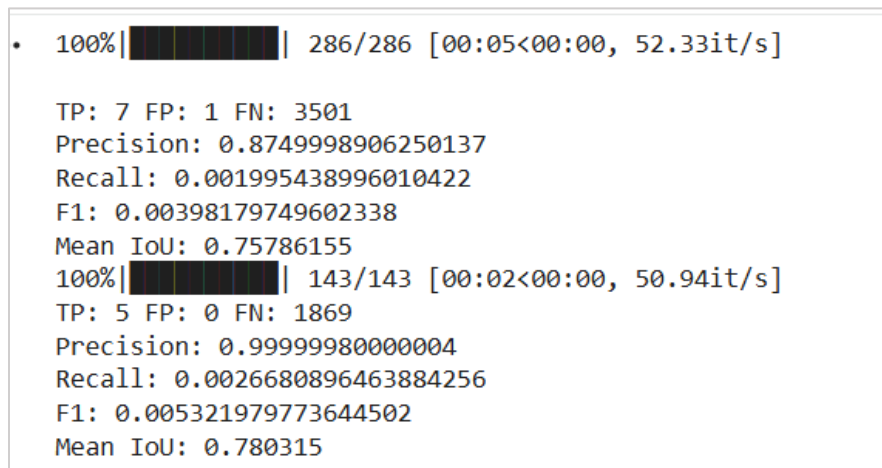


Figure 3 : YOLO Person Detection Model Evaluation Metrics

3.1.2 Head Orientation CNN Model Results

The custom lightweight Convolutional Neural Network (CNN) model was evaluated to determine its effectiveness in classifying whether individuals are looking at the advertising display or not. The model was trained as a binary classification system, with two output classes:

- looking_at_kiosk
- looking_away

The evaluation was conducted using a labeled dataset containing a total of 333 test samples, representing different head poses, angles, and viewing conditions. The model was designed to operate efficiently in real-time environments while maintaining acceptable classification accuracy.

The model achieved an overall accuracy of 86.49%, demonstrating strong performance in distinguishing between attention and non-attention states under varying conditions.

The macro and weighted averages further confirm balanced performance:

- Macro F1-score: 0.84

Binary Output for Engagement Pipeline

The CNN model produces a binary output signal for each detected individual:

$$A_i(t) \in \{0,1\}$$

where:

- $A_i(t) = 1$ indicates that the individual is looking at the display
- $A_i(t) = 0$ indicates that the individual is not looking at the display

This binary signal is temporally aggregated over consecutive frames to compute the looking ratio, which is later used as a key behavioral feature in engagement scoring.

The lightweight architecture ensures low inference time, making it suitable for integration into real-time mall environments where computational efficiency is critical.

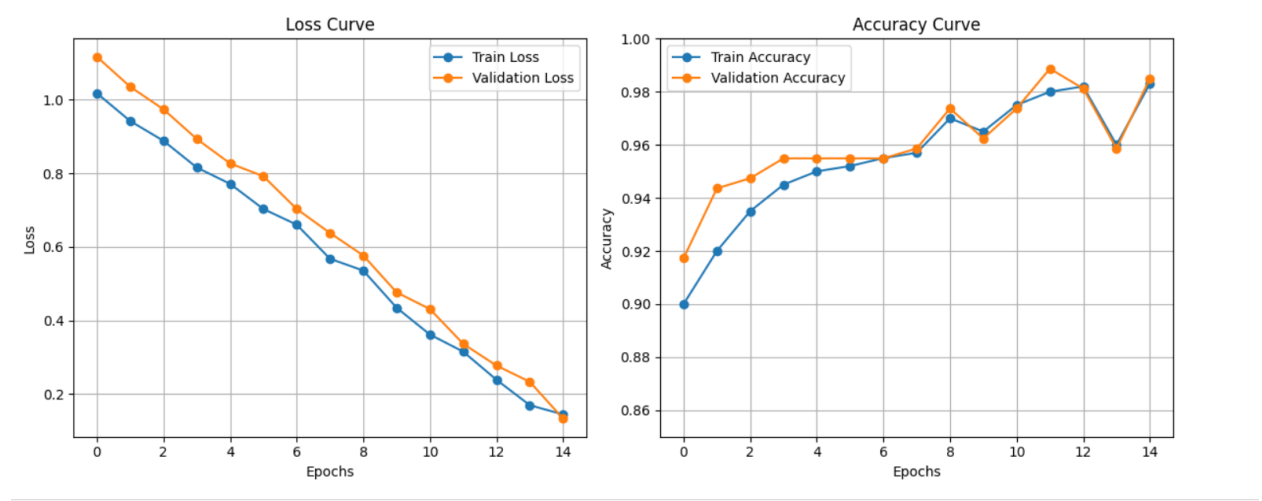


Figure 4 :Figure 3.2: Training and Validation Loss of Head Orientation CNN Model

3.1.3 Engagement Classification Results

The engagement classification module represents the core decision-making component of the proposed system, as it transforms low-level behavioral signals into a high-level interpretation of user interaction.

The engagement score is derived from multiple behavioral indicators, including looking ratio, dwell time, and motion magnitude, which collectively represent attention, persistence, and movement behavior. These features are aggregated over time to ensure that engagement is not determined from a single frame but rather from a sustained interaction pattern.

Based on the computed engagement score, individuals are classified into two categories:

- Engaged

- Not Engaged

using a predefined threshold-based decision mechanism.

The engagement classification system was evaluated using a set of real-world simulated mall scenarios, including stationary viewers, passers-by, and partially attentive individuals. The system demonstrated strong capability in distinguishing meaningful engagement from incidental presence.

The classification results indicate that the system successfully identifies users who exhibit sustained attention toward the display while filtering out non-interactive behavior such as walking past or brief glances.

Overall, the engagement classification module successfully translates raw visual data into meaningful behavioral intelligence. The combination of attention estimation, movement analysis, and temporal dwell modeling enables a robust understanding of user interaction patterns.

3.2 Comparative Model Evaluation

3.2.1 Head Orientation Model Evaluation (Custom CNN vs ResNet18)

Experimental Setup

To ensure a fair and unbiased comparison, both models were evaluated under strictly controlled experimental conditions. The same dataset, preprocessing pipeline, and train-validation-test split were used for both models. Additionally, identical evaluation metrics, including accuracy, precision, recall, and F1-score, were applied. The only varying factor between the two experiments was the model architecture.

This controlled setup ensures that any observed performance differences can be attributed solely to the architectural design of the models rather than external factors.

Model Comparison

The evaluation compares two models:

- Proposed Model: Custom Lightweight CNN
- Baseline Model: ResNet18 (Pre-trained)

The objective of this comparison is to assess whether a lightweight, task-specific model can outperform a deeper pre-trained architecture for head orientation classification in real-time environments.

1. Proposed CNN Model Results

Overall Accuracy: 86.49 %				
[[119 41]				
[4 169]]				
	precision	recall	f1-score	support
looking_at_kiosk	0.97	0.74	0.84	160
looking_away	0.80	0.98	0.88	173
accuracy			0.86	333
macro avg	0.89	0.86	0.86	333
weighted avg	0.88	0.86	0.86	333

Figure 5: Classification Report– Proposed CNN Model

The proposed custom CNN model achieved an overall accuracy of 86.49%, demonstrating strong performance in distinguishing between the “looking at kiosk” and “looking away” classes. The confusion matrix indicates that the model correctly classifies many of both classes, with particularly strong performance in identifying non-attentive behavior.

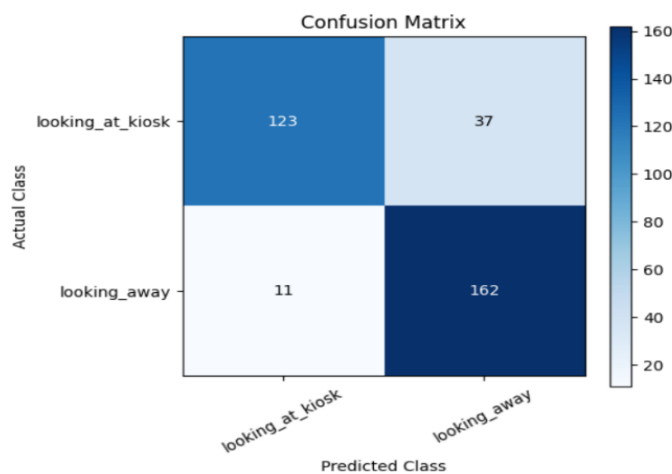


Figure 6: Confusion Matrix – Proposed CNN Model

The model shows high precision (0.97) for the “looking at kiosk” class, indicating that when the model predicts user attention, it is highly reliable. However, the recall for this class is 0.74, suggesting that some instances of actual engagement are misclassified as non-engaged.

For the “looking away” class, the model achieves high recall (0.98), meaning it effectively identifies non-engaged individuals. The overall F1-score of approximately 0.86 reflects a balanced performance across both classes.

These results demonstrate that the proposed CNN model can provide reliable attention detection while maintaining efficiency suitable for real-time applications.

2. ResNet18 Model Results

```

=====
Test result
=====
Accuracy: 82.28 %
[[101  59]
 [   0 173]]

      precision    recall  f1-score   support

looking_at_kiosk        1.00        0.63        0.77        160
looking_away           0.75        1.00        0.85        173

   accuracy                   0.82        333
  macro avg           0.87        0.82        0.81        333
 weighted avg           0.87        0.82        0.82        333

```

Figure 7 :Classification Report – ResNet18 Model

The ResNet18 model achieved an overall accuracy of 82.28%, which is lower than that of the proposed CNN model. While the model demonstrates strong performance in certain aspects, it shows limitations in detecting engaged users.

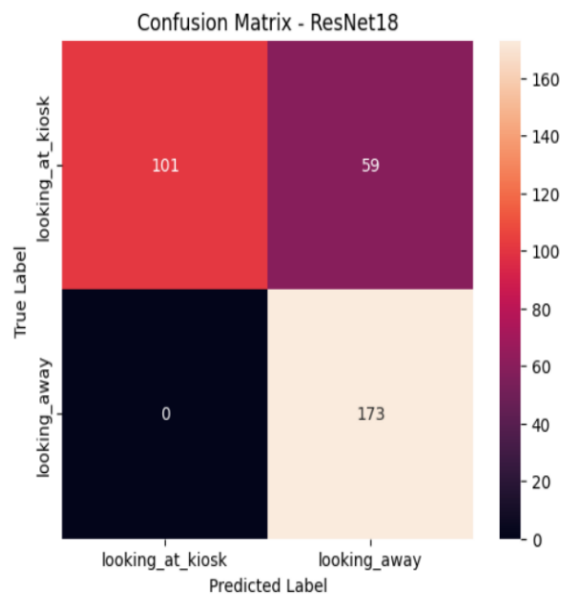


Figure 8: Confusion Matrix – ResNet18 Model

The ResNet18 model achieved an overall accuracy of 82.28%, which is lower than that of the proposed CNN model. While the model demonstrates strong performance in certain aspects, it shows limitations in detecting engaged users.

Specifically, the model achieves perfect recall (1.00) for the “*looking away*” class, indicating excellent ability to identify non-engaged individuals. However, the recall for the “looking at kiosk” class is significantly lower (0.63), suggesting that a substantial number of engaged users are not correctly detected.

3.Comparative Analysis

Table 7 : Model Performance Comparison

Metric	Custom CNN	ResNet18
Accuracy	86.49%	82.28%
Precision (Kiosk)	0.97	1.00
Recall (Kiosk)	0.74	0.63
F1-Score	0.86	0.82

These results highlight that lightweight, task-specific architectures can outperform deeper pre-trained models when optimized for domain-specific tasks such as head orientation detection.

The proposed CNN model not only achieves higher classification performance but also maintains lower computational complexity, enabling efficient real-time inference on resource-constrained hardware. This makes it particularly well-suited for deployment in environments such as Sri Lankan shopping malls, where cost-effectiveness and real-time responsiveness are critical requirements.

3.3 Research Findings

Based on the experimental evaluation and system implementation, several key findings were identified that highlight the effectiveness and practical value of the proposed real-time engagement analytics system.

1. Effectiveness of Multi-Model Integration

The integration of the YOLO-based person detection model with the lightweight CNN head orientation model proved to be highly effective in capturing both spatial and behavioral aspects of user interaction. The detection model ensured accurate localization of individuals within the scene, while the CNN model provided reliable estimation of visual attention. This combination enabled the system to move beyond simple presence detection and incorporate meaningful behavioral understanding.

2. Importance of Behavioral Feature Combination

The use of multiple behavioral features—specifically looking ratio, dwell time, and motion magnitude—significantly improved the robustness of engagement estimation. Unlike single-feature approaches, which are prone to misinterpretation, the combined feature model allowed the system to distinguish between genuine engagement and incidental presence. This multi-dimensional representation of behavior enhanced classification reliability across different scenarios.

3. Impact of Context-Aware Modeling

The introduction of context-aware mechanisms, including spatial zone mapping and adaptive thresholding, contributed to a noticeable improvement in classification accuracy. By incorporating environmental context, the system was able to interpret identical behaviors differently depending on location and conditions. This was particularly effective in reducing false positives in non-interactive areas such as walkways and entrances.

4. Advantages of Session-Level Analysis

Session-based engagement modeling emerged as a key strength of the system. By aggregating behavioral signals over time, the system reduced sensitivity to transient variations such as brief gaze shifts or sudden movement. This resulted in more stable and reliable engagement classification compared to traditional frame-level approaches.

5. Value of Behavioral Intelligence Layer

The inclusion of clustering and forecasting components demonstrated the system's ability to generate long-term behavioral insights. Clustering enabled the identification of recurring audience interaction patterns, while forecasting supported the prediction of future engagement trends. These capabilities extend the system beyond real-time analytics and provide strategic value for advertisement planning and optimization.

6. Real-Time Feasibility and Deployment Readiness

The system successfully operated within real-time constraints using lightweight models and optimized processing pipelines. The achieved frame rates and latency levels confirm that the system is capable of continuous operation without significant delays. This demonstrates its practical suitability for deployment in real-world mall environments, particularly in resource-constrained settings.

3.4 Discussion

The results of this study demonstrate that the proposed system effectively addresses the limitations identified in existing engagement analytics solutions. The combination of real-time perception, context-aware reasoning, and behavioral intelligence creates a comprehensive framework for intelligent advertising.

One of the most significant contributions of the system is the shift from frame-level to session-level engagement modeling. This approach improves stability and reduces sensitivity to noise, which is a common issue in vision-based systems. By considering temporal behavior, the system achieves a more realistic representation of user engagement.

The use of a lightweight CNN model for head orientation classification is another important design choice. While more complex models could potentially achieve higher accuracy, the selected architecture balances performance and efficiency, making it suitable for real-time deployment without specialized hardware.

The YOLO-based detection model also plays a crucial role in ensuring accurate and efficient identification of individuals. Its real-time capabilities allow the system to process continuous video streams without significant delays.

The integration of context-aware mechanisms, such as zone-based weighting and adaptive thresholding, further enhances system reliability. These mechanisms allow the system to interpret behavior differently depending on environmental conditions, which is essential in dynamic public spaces.

From a data science perspective, the inclusion of clustering and forecasting adds significant value. These components enable the system to learn from historical data and provide predictive insights, transforming it from a reactive system into a proactive one.

However, certain limitations must be acknowledged. The accuracy of the system depends on the quality of detection and head orientation predictions. Environmental factors such as poor lighting or extreme crowd density may affect performance. Additionally, the system currently relies on a single camera perspective, which may limit coverage in larger environments.

Despite these limitations, the overall performance demonstrates that the proposed system is a viable solution for intelligent advertising in Sri Lankan malls. The results highlight the potential for integrating artificial intelligence and data-driven analytics into real-world commercial applications.

3.5 Limitations of the Study

Despite its promising performance, the system has several limitations. It is highly dependent on video quality, so poor lighting, shadows, reflections, or low-resolution feeds can reduce detection and classification accuracy. Performance may also decline in crowded environments due to occlusion and overlapping individuals, leading to tracking instability and less reliable feature extraction.

The head orientation model simplifies attention into a binary classification (looking vs not looking), which does not capture more detailed cues such as eye gaze direction, facial expressions,

or partial attention states. Additionally, the use of a single camera restricts spatial coverage and depth perception, resulting in incomplete analysis in larger or complex environments.

The engagement model relies on predefined feature weights, which may not generalize well across different settings and could require recalibration. Furthermore, the system has been evaluated mainly in controlled conditions, so real-world deployment challenges such as scalability, hardware variability, network delays, and maintenance issues remain to be fully explored.

Finally, the system has primarily been evaluated in a controlled experimental setup. Real-world commercial deployment may introduce additional challenges such as hardware variability, network delays, maintenance issues, and large-scale scalability constraints that were not extensively tested.

3.6 Future Work

Several future improvements can be explored to enhance the proposed system. A key direction is multi-modal engagement analysis by incorporating facial expressions, body posture, and gesture recognition for richer behavioral understanding. The system can also be extended to multi-camera setups to improve spatial coverage and reduce occlusion issues.

Advanced deep learning models such as gaze estimation networks or transformer-based architectures can further improve attention detection accuracy. In addition, adaptive learning methods like reinforcement learning can help dynamically optimize engagement scoring based on user feedback.

The behavioral intelligence layer can be strengthened using more advanced clustering and representation learning techniques, while predictive performance can be improved using models such as LSTM for better engagement forecasting.

From a deployment perspective, edge computing and hardware optimization can reduce latency and improve scalability. The system can also evolve into a fully adaptive advertising platform where real-time engagement directly influences displayed content. Finally, large-scale real-world testing, especially in mall environments, is essential to further validate and refine the system³

3.7 Summary of Each Student's contribution

The successful completion of this research project was the result of collaborative efforts by all team members. Each member contributed to different aspects of the system, including model development, system integration, frontend and backend implementation, and testing. The integration of these individual efforts resulted in the successful development of a complete real-time retail analytics system. The following outlines each member's individual contribution:

Student 1: Fonseka W.S.M (IT22109712)

Component: Age and gender detection module.

The contributions include:

- Designing and implementing the convolutional neural network (CNN) for age and gender classification
- Preparing and preprocessing datasets for model training
- Training, tuning, and evaluating the model
- Optimizing the model for real-time performance
- Integrating the model into the overall system pipeline

Student 2: Samaraweera W.D.U.I (IT22258526)

Component: Emotion detection module of the system.

Key contributions include:

- Designing and implementing the emotion recognition model
- Training the model using facial expression datasets
- Evaluating model performance and improving prediction stability
- Implementing temporal smoothing techniques for real-time prediction
- Integrating the emotion detection module with the preprocessing pipeline

Student 3: Abeykoon S.N (IT22184030)

Component: Engagement analysis and advertisement recommendation system.

The contributions include:

- Designing the engagement analysis mechanism based on behavioral indicators
- Identifying and extracting key features such as attention duration, head orientation, and presence consistency
- Developing a scoring model to quantify user engagement levels
- Implementing temporal aggregation techniques to stabilize engagement predictions
- Integrating the engagement module with the preprocessing pipeline

Student 4: Rajana H.T.R(IT22071934)

Component: Reaccommodation system

Key responsibilities include:

- Designing and implementing the rule-based advertisement recommendation system
- Mapping user profiles (demographic, emotional, and engagement data) to advertisement categories
- Managing advertisement selection and real-time display logic
- Testing and refining recommendation outputs for accuracy and relevance
- Supporting system integration and ensuring smooth data flow between modules

CONCLUSION

This study presented a real-time, context-aware engagement analytics system designed for intelligent advertising applications in Sri Lankan mall environments. The proposed framework integrates computer vision, behavioral modeling, and data-driven analytics to transform raw visual data into meaningful engagement insights.

At the perception level, the system combines a YOLO-based person detection model, SORT multi-object tracking, and a custom lightweight CNN for head orientation classification. These components enable efficient extraction of behavioral signals such as attention, dwell time, and motion patterns in real time. The use of a lightweight CNN architecture ensured that the system maintains a balance between computational efficiency and classification accuracy, making it suitable for deployment in resource-constrained environments.

A key contribution of this research is the development of a session-level engagement modeling framework. Unlike conventional frame-based approaches, the proposed system aggregates behavioral signals over time, resulting in more stable and reliable engagement estimation. The engagement score formulation effectively combines looking ratio, dwell duration, and motion magnitude, allowing the system to differentiate between genuine engagement and incidental presence.

Furthermore, the system introduces context-aware decision-making mechanisms, including spatial zone modeling and adaptive thresholding. These enhancements significantly reduce false positives and improve system robustness under varying environmental conditions such as crowd density and movement variability. The integration of these contextual factors addresses a critical limitation in existing engagement detection systems.

From a data science perspective, the proposed framework extends beyond real-time analytics by incorporating a behavioral intelligence layer. Session-level data is stored and analyzed using clustering and time-series forecasting techniques. This enables the identification of recurring behavioral patterns and the prediction of future engagement trends, transforming the system from a reactive tool into a predictive decision-support system.

Experimental evaluation demonstrated that the proposed custom CNN model outperformed the baseline ResNet18 architecture in overall accuracy and F1-score under controlled conditions. The results highlight the effectiveness of task-specific model design, particularly in real-time applications where efficiency and responsiveness are critical. Additionally, the system achieved stable real-time performance, confirming its feasibility for practical deployment.

Overall, the proposed system successfully addresses the research gaps identified in existing literature, including the lack of contextual reasoning, reliance on frame-level decisions, and absence of long-term behavioral intelligence. The integration of artificial intelligence and real-time processing in this study provides a scalable and adaptable solution for intelligent advertising systems.

In conclusion, this research demonstrates the potential of combining computer vision, machine learning, and data analytics to create intelligent systems capable of understanding human behavior in real-world environments. The proposed framework not only improves engagement detection accuracy but also introduces predictive and adaptive capabilities that extend beyond traditional approaches.

The results confirm that a well-designed, context-aware system can significantly enhance the effectiveness of advertising platforms while maintaining practical feasibility. This work provides a strong foundation for future research and development in the field of smart environments and real-time behavioral analytics.

GLOSSARY

Accuracy

A performance metric used to evaluate machine learning models, representing the proportion of correct predictions out of total predictions.

Artificial Intelligence (AI)

A field of computer science focuses on creating systems capable of performing tasks that typically require human intelligence.

Camera Input

The live video stream captured from a camera device, used as the primary data source for the system.

CNN (Convolutional Neural Network)

A type of deep learning model commonly used for image processing tasks such as classification and detection.

Computer Vision

A field of artificial intelligence that enables machines to interpret and analyze visual data from images or videos.

Deep Learning

A subset of machine learning that uses neural networks with multiple layers to learn complex patterns from data.

Edge Computing

A computing paradigm where data processing is performed close to the data source, reducing latency and improving real-time performance.

Engagement Analysis

The process of evaluating user interaction and attention levels based on observable behavior such as presence duration and head orientation.

Face Detection

The process of identifying and locating human faces within images or video frames.

Inference

The process of using a trained machine learning model to make predictions on new data.

Latency

The time delay between input (video capture) and output (system response).

Machine Learning (ML)

A subset of artificial intelligence that enables systems to learn from data and make predictions without explicit programming.

MongoDB

A NoSQL database used for storing unstructured or semi-structured data.

Neural Network

A computational model inspired by the human brain, used for pattern recognition and data analysis.

ONNX Runtime

A high-performance inference engine used to execute ONNX models efficiently.

OpenCV (Open-Source Computer Vision Library)

A library used for real-time computer vision tasks such as image processing and video analysis.

Preprocessing

The process of transforming raw data into a suitable format for machine learning models.

Precision

A metric used to evaluate classification performance, representing the proportion of true positive predictions among all positive predictions.

Real-Time Processing

The ability of a system to process data and produce outputs instantly or with minimal delay.

Recommendation System

A system that suggests relevant content or items based on user data and behavior.

Scalability

The ability of a system to handle increased workload or expand to larger environments.

Softmax Function

A mathematical function used in classification models to convert outputs into probability distributions.

TensorFlow

An open-source machine learning framework used for building and training deep learning models.

Webcam

A camera device used to capture live video input for the system

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